# Jake Hobson

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## Education

**University of California, Santa Cruz** — Santa Cruz, CA
*B.S. Robotics Engineering, Minor in Electrical Engineering | GPA: 3.84/4.00* — *Sep 2021 – Jun 2026*

- Honors: Dean's Honors (top 15%, 2021–2025); Tau Beta Pi Engineering Honor Society

## Experience

**Project Control Systems Lead** — Nov 2025 – Jun 2026
*University of California, Santa Cruz* — *Santa Cruz, CA*

- Led the controls subsystem as one of four team leads (mechanical, power, systems integration) and served as the team's finance lead, managing the project budget.
- Designed and built a 1:10-scale proof-of-concept surfboard mechanical testing rig to quantify flexibility, strength, and load response.
- Programmed microcontroller-based actuator control systems capable of applying loads up to 150N with less than 1% error.
- Developed a user interface for real-time data visualization and control of test parameters.

**Mechanical Engineering Intern** — Jun 2023 – Oct 2023
*Bedrock Ocean Exploration* — *Richmond, CA*

- Reduced AUV recovery time by 35% by designing, fabricating, and testing a new latch mechanism for the Launch and Recovery System (LARS).
- Migrated and refactored CAD assemblies from SolidWorks to Autodesk Inventor, resolving compatibility issues and redesigning critical structures to improve robustness.
- Partnered with senior engineers to validate mechanical designs through hands-on prototyping and iterative testing.

## Projects

**Throwing Trajectory Estimator** | *Python, C* — Feb 2026 – Mar 2026

- Partnered with two teammates to engineer a wearable inertial sensing system that estimates 3D projectile trajectories for throwing and pitching applications.
- Applied 9-DOF IMU sensor fusion with bias-drift correction to estimate release velocities and angular velocities.
- Created a Python simulation that projected baseball horizontal displacement within 10% of measured values, using release velocity vectors.

**Underwater Propulsion Model** | *Python* — Feb 2026 – Mar 2026

- Worked alongside three teammates to construct a physics-based underwater vehicle propulsion simulator using hydrodynamic models derived from Fossen (1994).
- Evaluated a conventional propeller model against a novel jellyfish-inspired "umbrella" propulsion mechanism.

**Optimizing Traffic Lights as a Cyber-Physical System** | *MATLAB/Simulink* — Nov 2025 – Dec 2025

- Coordinated with four teammates to model a traffic intersection as a hybrid cyber-physical system, integrating continuous vehicle-flow dynamics with a discrete finite-state-machine controller.
- Optimized control logic combining queue-size sensing and timers, reducing average wait times by ~30% (peak reduction ~64%) across tested traffic arrival rates.

**Autonomous Maze-Runner Robot** | *C* — May 2025 – Jun 2025

- Teamed up with two classmates to assemble an autonomous mobile robot that navigates a maze, detects and retrieves a flag, and returns it to a raised platform within a 4-week development cycle.
- Delivered the complete system for $87.13, 42% under the $150 project budget.
- Coded embedded firmware for line-following and actuator control under strict budget constraints.

## Skills

**Programming**: Python, MATLAB/Simulink, C/C++, ROS/ROS2
**Controls**: PID Control, Actuator Control Systems, Sensor Fusion, State Estimation
**Electrical & Hardware**: Embedded Systems (STM32, PIC32, Raspberry Pi Pico), Signal Processing, Communication Protocols (I2C, SPI, UART)
**Software & Tools**: Git, LaTeX, Linux
**Mechanical**: CAD (Onshape, SolidWorks, Autodesk Inventor), FEA/Structural Simulation, Prototyping & Fabrication
**Soft Skills**: Subsystem Ownership, Cross-functional Team Coordination, Project Management
**Languages**: English (Native), Japanese (Elementary)